



AI Engineer Take-Home Exercise

Build a Customer Intent Triage Agent

The setup

You've been handed a folder of materials from a fictional client — **Northwind Home Services**, a residential trades business. Their support inbox is a mess. Customer enquiries land in one channel, and a human triages them by hand against an ageing SOP. They want an agent that does the first pass.

Your job is to build that agent. We've given you everything a junior consultant on their first day would receive: an SOP, a service catalogue, a tone guide, and a batch of real-looking inbound messages. We've also given you a benchmark — a gold-standard set of how the agent *should* have handled those messages — so you can score your own work.

What you'll build

An agent (single agent or multi-agent — your call) that takes one inbound customer message and produces a structured triage decision. You'll run it across the 20-message dataset, compare your output to the benchmark, and report your accuracy.

For each inbound message, your agent should output:

Field	Description
category	One of: BOOKING, QUOTE, COMPLAINT, EMERGENCY, BILLING, OUT_OF_SCOPE
priority	P1 (same-day), P2 (within 48h), P3 (within 5 business days)
route_to	Which team handles it (per the SOP)
draft_reply	A short customer-facing reply, following the tone guide
needs_human_review	true / false — flag anything ambiguous or outside policy
reasoning	1–3 sentences on why you made these calls

What's in the pack

File	What it is
01_Candidate_Brief.pdf	This document.
02_Northwind_SOP.pdf	How the business currently handles inbound enquiries. The agent's rulebook.
03_Service_Catalogue.pdf	Services the business offers, with pricing tiers and constraints.
04_Tone_and_Style_Guide.pdf	How Northwind speaks to customers. Your draft replies need to match this.

05_Inbound_Messages.json	20 customer messages. The input to your agent.
06_Benchmark.json	Gold-standard triage decisions for those 20 messages. Your scorecard.
07_Evaluation_Rubric.pdf	How to grade your agent against the benchmark.

How we're thinking about this

Three things, roughly in this order:

1. Did you actually read the materials? The SOP, catalogue, and tone guide each contain rules that interact. A good answer needs all three. Some inbound messages are deliberately ambiguous or cross-cutting — there are cases where the SOP and the catalogue disagree, and cases where neither covers what the customer is asking. We want to see how you handle that.

2. Is the agent design sensible? We don't care which framework, which model, or which orchestration pattern you use. We care that the choices you made are defensible — and that you can tell us why. A well-prompted single agent can absolutely beat a poorly-designed multi-agent system on a task this size. Pick what fits.

3. Did you grade yourself honestly? The benchmark isn't perfect. There are messages where your agent might reasonably disagree with our gold answer — we want to know when you spotted that, and what you'd change about the benchmark itself. An agent that scores 80% with sharp self-critique beats one that scores 95% and shrugs.

What to send back

A short write-up (markdown, doc, slides — whatever) and your code. Aim for:

- Your agent design in a paragraph or two — what it does, why that shape.
- The prompts you ended up with (final versions, no need to show every iteration).
- Your accuracy score against the benchmark, broken down by field.
- 3–5 cases where you disagreed with the benchmark, or where the agent struggled, with your read on why.
- One thing you'd build next if you had another day.

Time budget: 1–2 hours. We know your time is valuable, and we've scoped this exercise so you don't need more than that to show us what we need to see. If you're three hours in, please stop and write up what you have — a partial submission with clear thinking is more useful to us than a polished one that ate your weekend.

Ground rules

- Use any model, framework, or library you like.
- Use AI to help you build it. We'd be surprised if you didn't.
- Don't hand-tune for the benchmark — build the agent, then run it. If you spot a benchmark you disagree with, note it; don't quietly correct your agent to match.
- If something in the materials is unclear or contradictory, that's the exercise. Make a call and tell us what you decided.

Good luck. We're more interested in how you think than how polished it looks.